

Project Close-out Report

ZK from Zero on Cardano: eBook

Catalyst Fund 14 • Cardano Open: Ecosystem • Project #1400129 • Raul Rosa (elRaulito) • 60,000 ADA • 8 months • AFL-3.0

Close-out video: youtu.be/FS40K_45CWQ **Repository:** github.com/elRaulito/ZK-from-zero-on-Cardano
Proposal: projectcatalyst.io/funds/14/cardano-open-ecosystem/zk-from-zero-on-cardano-ebook-by-elraulito

What Was Delivered

A free, open-source, 107-page technical book taking a developer from the mathematics of zero-knowledge proofs to a working, tested ZK smart contract on Cardano, plus a runnable reference implementation. Nine chapters and a sitography of the whole ZK-on-Cardano landscape. Built reproducibly in LaTeX via Nix and GitHub Actions, in light and dark themes.

- **Book:** `eBook/light.pdf` and `eBook/dark.pdf`, 107 pages each, published to the `rendered` branch on every push.
- **Reference implementation:** `zk-password/`, a password-locked UTxO unlocked by a Groth16 proof. Circom circuits, snarkjs trusted setup, point compression, Aiken verifier and validator, 15 passing tests, on-chain cost benchmarks.

KPIs and Milestones

Project KPI	Target	Status
Chapters written	9	Complete
Working ZK contract	1	Complete
Circom circuits	1+	Complete (2)
Off-chain pipeline docs	Yes	Complete
Advanced ZK case study	1	Complete
Reproducible build	Yes	Complete
Open-source licence	Yes	Complete
Milestone reports	4	Complete
Close-out report + video	1 each	Complete

Milestone	Content	Done
M1	Ch. 1–2: foundations, history	Yes
M2	Ch. 3–4: applications, privacy	Yes
M3	Ch. 5–7: contract, off-chain	Yes
M4	Ch. 8–9: case study, ethics	Yes

All challenge KPIs met: educational material, lowered barrier to entry, reusable open-source artifacts, original technical findings, and free permissionless access.

Key Achievements: Measured, Not Asserted

The project's most valuable output is a set of numbers nobody had published. All are reproducible with `aiken check -m bench` in `zk-password/`.

Finding	Value	Why it matters
Compiled Groth16 verifier	570 bytes	The on-chain part is tiny
Cost to verify a proof	2,005,373,690 steps	≈ 0.147 ADA, 20% of the CPU budget
... and that cost is	constant	Independent of circuit size
Max verifications per tx	4	Hard ceiling, CPU-bound
Cost floor per Plutus operation	≈ 2 × 10 ⁵ steps	Interpreter overhead, not operand width
On-chain Poseidon vs blake2b	≈ 1,500×	No optimisation escapes the floor
Depth-20 Poseidon Merkle insert	120% / 173%	Of CPU / memory budget: does not fit

Headline conclusion. Cardano can verify a zero-knowledge proof about any computation, of any size, for about 0.147 ADA and 570 bytes of on-chain code. It cannot yet maintain the kind of cryptographic state the most ambitious ZK applications need, because it lacks a ZK-friendly hash builtin and the eUTxO ledger deliberately has no shared mutable state. **A single Poseidon builtin is the highest-leverage fix available.**

Chapter 8 applies this to a Tornado Cash design and reaches a clearly argued negative result, identifying three walls: no ZK-friendly hash builtin; a hash-function dilemma where the on-chain-cheap hash forces server-side proving and therefore destroys the privacy; and eUTxO contention, where a pool's anonymity set is capped by its throughput of one deposit per block. It also documents where eUTxO beats Ethereum: minting a token named after the nullifier

hash makes double-spend prevention a ledger invariant, not a script check.

Key Learnings

- **The on-chain part is the small part.** A ZK application on Cardano is an off-chain application with a 570-byte on-chain referee. Off-chain logic needed a full chapter, not the half we budgeted.
- **Cardano’s ZK cost profile is bimodal, not graded.** Cryptography with a builtin is nearly free; cryptography without one is effectively unavailable. No amount of careful Aiken bridges that gap.
- **Measuring changed our conclusions.** The original outline assumed “too expensive, roughly.” Measurement located the cost in interpreter overhead rather than bignum width, which rules out whole classes of proposed optimisation. Budget time for measurement, not only exposition.
- **Negative results are worth publishing.** Documenting why something cannot be built, with numbers, saves other teams weeks.

Next Steps

- Add replay protection to the reference contract: a nonce in the datum and the circuit’s public inputs. An afternoon’s work, left as a reader exercise.
- Implement Poseidon in Aiken and publish real measurements, replacing our labelled lower-bound extrapolations with data.
- Argue for a Poseidon builtin using these numbers, published under AFL-3.0 precisely so they can be reused in a CIP or governance thread.
- Build with off-chain computation compression: constant verification cost means an arbitrarily large computation verifies for ≈ 0.147 ADA, with none of Chapter 8’s walls. The best effort-to-value ratio available on Cardano today, and badly underexploited.

The repository stays open for issues and pull requests. Corrections are more welcome than praise, and the Chapter 8 estimates in particular deserve to be replaced by someone’s real implementation.

Links

This project

Close-out video	youtu.be/FS40K_45CWQ
Repository	elRaulito/ZK-from-zero-on-Cardano
Proposal	Catalyst Fund 14, #1400129
Benchmarks	<code>aiken check -m bench</code>
Author	raul.it

Prior work credited in the book

Janus Wallet	leobel/janus-wallet
aiken-zk (Eryx)	eryxcoop/cardano-zk-aiken
Groth16 in Aiken	Modulo-P/ak-381
Semaphore port	Modulo-P/Cardano-Semaphore
BLS12-381 gym	cardano-foundation/bls

Written by Raul Rosa ([elRaulito](https://github.com/elRaulito)). Reviewed with [Claude Code](#). Typeset in LuaLaTeX. Released under AFL-3.0.
